GAP-10D Endgame Audit: Axiomatic Underdetermination and the Correct Induced-Einstein Coefficient

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Abstract

We determine exactly how far the present UBT axioms can derive Einstein dynamics. First, we prove an underdetermination theorem: the covariant-tetrad map, central metric identity, split-jet right inverse, and all currently stated local symmetries do not fix the coefficient of a curvature action. Infinitely many actions with different Newton constants share the same UBT kinematics. Thus an unconditional derivation of G from the existing kinematic axioms is logically impossible. Second, we compute the curvature term generated by a proper-time regulated one-loop determinant of a specified Laplace-type Θ Hessian on $M^4 \times S^1_\psi$. A companion principal-symbol audit now proves that the fixed-background kinetic Hessian is of this scalar Laplace type, while the six-dimensional $q = 1$ D-composite resonance is not a conic principal-symbol bundle. The result below gives an exact conditional formula for the induced Newton coefficient, including the full Kaluza–Klein heat trace. At the self-dual cutoff $\Lambda R_\psi = 1$ the dimensionless KK factor is $1.303410251859\ldots$. This closes the form of the induced-Einstein bridge, but the physical mode count, nonminimal curvature coupling, and cutoff identification must be derived or explicitly adopted before the numerical Newton constant is a UBT prediction.

1 What the current axioms do not determine

Let $g[\Theta]$ denote any metric admitted by the local split-jet construction. The current geometric axioms specify the field space and the map to the metric, but not a unique scalar functional of that metric.

Theorem 1 (Curvature-coefficient underdetermination). *For every real constant c, the action family*

$$S_c[\Theta] = S_{\text{locked}}[\Theta] + c \int d^4x \, \sqrt{-g[\Theta]} \, R[g[\Theta]] \tag{1}$$

has the same covariant-tetrad definition, central metric identity, local Lorentz symmetry, split-jet representability, and kinematic connection reconstruction. Distinct values of c give distinct Newton constants. Therefore those kinematic statements cannot determine c.

Proof. Every item listed in the theorem is a definition or an algebraic/differential identity on the field space and is independent of the coefficient multiplying a separately invariant scalar functional. The second term in Eq. (1) is diffeomorphism and local-Lorentz invariant for every c. Changing c leaves all kinematic identities unchanged but changes the metric Euler–Lagrange equation and the Newton coefficient. Hence c is not a logical consequence of the current kinematic axioms. □

The same argument applies to R^2, $R_{\mu\nu}R^{\mu\nu}$, and other higher- derivative invariants. Lovelock uniqueness identifies the two-derivative infrared form once its assumptions are adopted; it does not determine its overall coefficient or derive the action from Θ.

Corollary 2 (Absolute-scale obstruction)**.** *The present normalization convention treats $\mathcal{N}_0$ as a global unit-setting constant. Under a change of mass unit by a positive factor a,*

$$\mathcal{N}_0 \mapsto a^4\mathcal{N}_0, \qquad \Lambda \mapsto a\Lambda, \qquad G \mapsto a^{-2}G, \tag{2}$$

while all dimensionless geometric identities are unchanged. Consequently an absolute numerical value of the dimensionful constant G cannot follow from the scale-free kinematic identities alone. The meaningful candidate prediction is a dimensionless ratio such as $G\sqrt{\mathcal{N}_0}$; even that ratio requires the physical Hessian, measure and matching prescription.

2 Principal-symbol decision

The companion note `gap_10d_theta_hessian_principal_symbol.tex` separates two operators that had previously been discussed together. For fixed $g_{\mu\nu}$, background connection and nondegenerate internal pairing, the raised Hessian of the quadratic Θ kinetic term obeys

$$\sigma_2(\mathcal{H}^A{}_B)(x,k) = g^{\mu\nu}k_\mu k_\nu\,\delta^A{}_B.$$

Thus the operator-class assumption used below is proved for the fixed-background fluctuation problem. The frozen D-composite full symbol $I - A(s,\lambda)$ is different: its six-dimensional singular sector occurs at $\lambda \cdot s = 1$, which is not invariant under radial rescaling of s. It therefore is not a homogeneous principal-symbol bundle and cannot be inserted directly as six physical heat-kernel modes. The full composite, gauge-fixed Θ Hessian, including all variations of $E[\Theta]$, $g[\Theta]$ and $\Omega[\Theta]$, remains the decisive missing operator.

3 Specified one-loop branch

A genuine induced-gravity calculation must begin with a specified quadratic fluctuation operator, not merely an abstract Dirac operator; see the standard heat-kernel review [2] and the induced-gravity proposal of Sakharov [1]. Assume that after background expansion, gauge fixing, and removal of nonphysical directions the Euclidean Hessian contains N_B real bosonic modes with Kaluza–Klein label $n \in \mathbb{Z}$ and Laplace-type operators

$$P_n = -\nabla^2 + m^2 + \frac{n^2}{R_\psi^2} + \xi R + \mathcal{E}_0, \tag{3}$$

where $\mathcal{E}_0$ has no term proportional to the scalar curvature. The one-loop contribution is

$$\Gamma_1 = \frac{1}{2}\sum_n \operatorname{Tr}\log P_n = -\frac{1}{2}\int_{\Lambda^{-2}}^{\infty}\frac{ds}{s}\sum_n \operatorname{Tr} e^{-sP_n}, \tag{4}$$

using a proper-time cutoff Λ.

For the four-dimensional heat kernel,

$$\operatorname{Tr}_{M^4} e^{-sP_n} = \frac{N_B}{(4\pi s)^2}\int d^4x\sqrt{g}\,e^{-s(m^2+n^2/R_\psi^2)}\left[1 + s\left(\frac{1}{6}-\xi\right)R + O(s^2)\right]. \tag{5}$$

Define the exact regulated KK integral

$$\mathcal{I}_1(\Lambda, R_\psi, m) := \int_{\Lambda^{-2}}^{\infty} ds\, s^{-2} e^{-m^2 s}\vartheta_3\Big(0, e^{-s/R_\psi^2}\Big). \tag{6}$$

Theorem 3 (Induced Einstein coefficient). *Under the stated Hessian and regulator assumptions, the curvature part of the one-loop Euclidean action is*

$$\boxed{\Gamma_1\big|_R = -\frac{N_B(1-6\xi)}{192\pi^2}\,\mathcal{I}_1(\Lambda, R_\psi, m)\int d^4x\sqrt{g}\,R.} \tag{7}$$

Matching the Euclidean convention $S_E = -\frac{1}{16\pi G}\int\sqrt{g}\,R + \cdots$ gives

$$\boxed{\frac{1}{G_{\rm ind}} = \frac{N_B(1-6\xi)}{12\pi}\,\mathcal{I}_1(\Lambda, R_\psi, m).} \tag{8}$$

For minimally coupled bosonic modes, $\xi = 0$, the induced Newton constant has the physical sign.

Proof. Insert Eq. (5) into Eq. (4). The coefficient of R is

$$-\tfrac{1}{2}(4\pi)^{-2}N_B(\tfrac{1}{6}-\xi)\mathcal{I}_1,$$

which is Eq. (7). Matching coefficients yields Eq. (8). □

4 Low-energy and self-dual limits

If $\Lambda R_\psi \ll 1$ and $m \ll \Lambda$, nonzero KK modes are exponentially suppressed and

$$\mathcal{I}_1 = \Lambda^2 + O(e^{-1/(\Lambda R_\psi)^2}). \tag{9}$$

At the self-dual identification $\Lambda R_\psi = 1$ and $m = 0$, put $s = u/\Lambda^2$. Then

$$\mathcal{I}_1 = C_\psi \Lambda^2, \qquad C_\psi := \int_1^\infty \frac{du}{u^2}\vartheta_3(0, e^{-u}). \tag{10}$$

Termwise integration gives the rapidly convergent exact representation

$$C_\psi = 1 + 2\sum_{n=1}^{\infty}\left[e^{-n^2} - n^2 E_1(n^2)\right] = 1.303410251859279308\ldots. \tag{11}$$

Therefore

$$\frac{1}{G_{\rm ind}} = \frac{N_B(1-6\xi)C_\psi}{12\pi}\Lambda^2. \tag{12}$$

Dimensional analysis of $E_\mu = \mathcal{N}_0^{-1/2}D_\mu\Theta$ with $[\Theta] = 1$ gives $[\mathcal{N}_0] = 4$. If one explicitly adopts the scale-identification axiom

$$\Lambda = \mathcal{N}_0^{1/4} = R_\psi^{-1}, \tag{13}$$

then the reduced Planck mass obeys

$$\boxed{\bar{M}_{\rm Pl}^2 := \frac{1}{8\pi G_{\rm ind}} = \frac{N_B(1-6\xi)C_\psi}{96\pi^2}\sqrt{\mathcal{N}_0}.} \tag{14}$$

For the unconstrained eight-real-component biquaternion count and $\xi = 0$, the conditional numerical relation is

$$\bar{M}_{\rm Pl} = 0.10490594\,\mathcal{N}_0^{1/4}.$$

The physical gauge-fixed mode count need not equal eight.

5 Why the earlier five-dimensional note is not a closure proof

The exploratory May 2026 file `research_tracks/T3_ALPHA/seeley_dewitt_coefficients.tex` cannot be used as a GR coefficient derivation:

1. it introduces a spectral Dirac operator that is not shown to be the quadratic Hessian of the canonical Θ action;
2. it double-counts the Lichnerowicz $R/4$ term by writing it once inside $D^2_{M^4}$ and once again in the product operator;
3. it mixes real and complex fibre dimensions in the trace count;
4. a five-dimensional local a_2 coefficient and a four-dimensional low-energy Newton coefficient have different cutoff scaling;
5. the finite KK theta factor belongs to the regulated full heat trace Eq. (6), not to a local Seeley–DeWitt coefficient by itself.

The corrected computation is Eq. (8).

6 Closure boundary

The induced formula is exact once (P_n, N_B, ξ, Λ) are specified. The current repository does not yet derive all four from the finalized constrained Θ measure. In particular, GAP-Q still lists the functional measure, gauge/ghost quotient, and UV scale as open. Therefore one may adopt a well-defined *spectral-induced effective branch*, but one may not relabel it as an unconditional consequence of the earlier kinematic axioms.

Exact status

GAP-10D-UNDERDETERMINATION: CLOSED AS NO-GO [L1]. The existing geometric axioms do not determine a curvature coefficient or Newton constant; an additional dynamical/quantum principle is necessary. Corollary 2 separately shows that the absolute dimensionful scale cannot be predicted before the physical scale represented by $\mathcal{N}_0$ is fixed.

GAP-10D-A2-FORM: CLOSED CONDITIONALLY [L1]. For a specified gauge-fixed Laplace-type Θ Hessian, proper-time one-loop integration generates the Einstein–Hilbert term with the exact coefficient Eq. (8), including the full compact-ψ KK trace.

GAP-10D-SPECTRAL-IR: CLOSED CONDITIONALLY [L1]. If the Hessian, physical mode count, curvature coupling, and cutoff/scale identification are adopted or independently derived, the two-derivative infrared metric action is Einstein–Lambda, with higher-curvature terms starting at the next heat-kernel order.

GAP-10D-HESS-FIXED: CLOSED [L1]. The fixed-background Θ kinetic Hessian has scalar Laplace-type principal symbol after raising the internal pairing index.

GAP-10D-RES-PRINCIPAL: CLOSED AS NO-GO [L1]. The six-dimensional $q = 1$ D-composite resonance is a nonconic finite-scale singularity of a mixed-order full symbol, not a principal-symbol Laplace bundle.

GAP-10D-HESS-COMP: NARROWED, not closed. The remaining operator lemma is the complete gauge-fixed composite Θ Hessian and its physical/ghost quotient.

GAP-10D: NARROWED, not unconditionally closed. The remaining first-principles work is derivation of that composite Hessian, (N_B, ξ, Λ) and the constrained fluctuation measure from the finalized UBT action, without fitting G.

References

[1] A. D. Sakharov, "Vacuum quantum fluctuations in curved space and the theory of gravitation," *Dokl. Akad. Nauk SSSR* **177**, 70–71 (1967) [*Sov. Phys. Dokl.* **12**, 1040–1041 (1968)].

[2] D. V. Vassilevich, "Heat kernel expansion: user's manual," *Phys. Rept.* **388**, 279–360 (2003).